

Surface reconstruction from scattered data by THB-splines for industrial applications

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International Workshop
on Numerical Mathematics and its Applications
Torino, 21 Settembre 2018

Outline of the talk

- ▶ Hierarchical spline spaces
 - ▶ Hierarchical meshes
 - ▶ THB-splines: a basis for hierarchical spline spaces
 - ▶ Hierarchical quasi-interpolants (QIs)
- ▶ Scattered data fitting by hierarchical splines
 - ▶ Local least squares approximation by variable degree polynomials
 - ▶ Scattered data hierarchical QI
 - ▶ Numerical tests

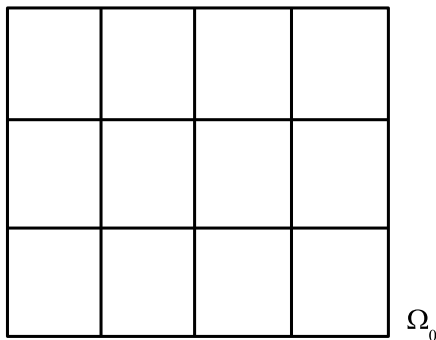
Hierarchical spline spaces

Hierarchical spline spaces - concept

Tensor-product spline space V^0 on the $N_x^0 \times N_y^0$ mesh $\mathcal{G}^0$, with corresponding basis

$$\mathcal{B}_{\mathbf{d}}^0 := \{B_J^0, J \in \Gamma_0\}, \quad (\text{degree } \mathbf{d} = (d_x, d_y)),$$

$$\Gamma_0 := \{(i, j) : i = -d_x, \dots, N_x^0 - 1, j = -d_y, \dots, N_y^0 - 1\}.$$



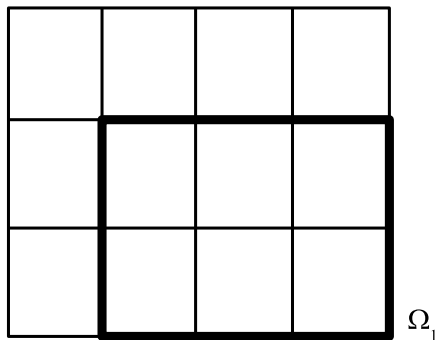
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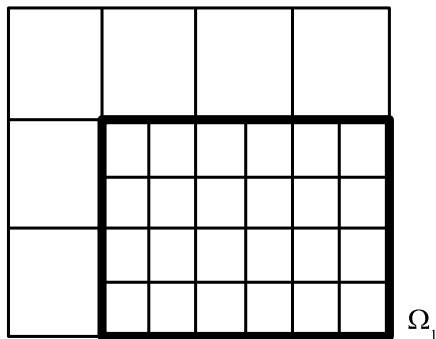
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Hierarchical spline spaces - concept

Tensor-product spline space V^1 on the $N_x^1 \times N_y^1$ mesh $\mathcal{G}^1$, with corresponding basis

$$\mathcal{B}_{\mathbf{d}}^1 := \{B_J^1, J \in \Gamma_1\}, \quad (\text{degree } \mathbf{d} = (d_x, d_y)),$$

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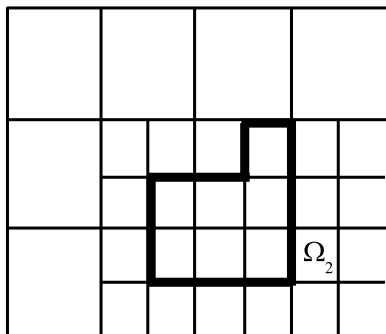
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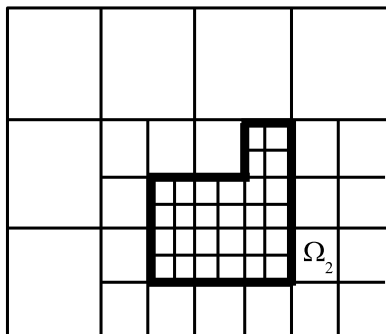
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Tensor-product spline space V^2 on the $N_x^2 \times N_y^2$ mesh $\mathcal{G}^2$, with corresponding basis

$$\mathcal{B}_{\mathbf{d}}^2 := \{B_J^2, J \in \Gamma_2\}, \quad (\text{degree } \mathbf{d} = (d_x, d_y)),$$

$$\Gamma_2 := \{(i, j) : i = -d_x, \dots, N_x^2 - 1, j = -d_y, \dots, N_y^2 - 1\},$$



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Hierarchical spline spaces

- ▶ sequence of nested tensor-product spline spaces on Ω

$$V^0 \subset V^1 \subset V^2 \subset \dots \subset V^{M-1}$$

with basis $\mathcal{B}_{\mathbf{d}}^{\ell} := \{B_J^{\ell}, J \in \Gamma_{\ell}\}, \quad \ell = 0, 1, 2, \dots, M-1,$

- ▶ sequence Ω_M of nested closed subsets of Ω

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We define the hierarchical B-splines basis:

$$\mathcal{B}_{\mathcal{G}_H} := \{B_J^\ell : J \in \Gamma_\ell : \text{supp}(B_J^\ell) \subseteq \Omega^\ell \wedge \\ \text{supp}(B_J^\ell) \not\subseteq \Omega^{\ell+1}, \ell = 0, \dots, M-1\}.$$

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Truncated Hierarchical B-splines (THB-splines) - definition

THB-splines were introduced in [Giannelli, Jüttler, Speleers; 2012]

For any $s \in V^\ell \subset V^{\ell+1}$, with $s = \sum_{J \in \Gamma_\ell} c_{J,\ell+1} B_J^{\ell+1}$, we set

$$\textcolor{red}{trunc}^{\ell+1}(s) := \sum_{\textcolor{red}{J:supp}(B_J^{\ell+1}) \not\subseteq \Omega^{\ell+1}} \textcolor{red}{c}_{J,\ell+1} B_J^{\ell+1},$$

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The basis $\mathcal{T}_{\mathcal{G}_\mathcal{H}}$ of THB-splines is defined by:

$$\mathcal{T}_{\mathcal{G}_\mathcal{H}} := \{ T_J^\ell = \textcolor{red}{Trunc}^{\ell+1}(B_J^\ell) : B_J^\ell \in \mathcal{B}_{\mathcal{G}_\mathcal{H}} \}.$$

EACH T_J^ℓ CORRESPONDS TO A B_J^ℓ

Hierarchical quasi-interpolants

We consider a **sequence of tensor-product quasi-interpolants**:

$$Q^\ell(f) := \sum_{J \in \Gamma_\ell} \mu_J^\ell(f) B_J^\ell, \quad \ell = 0, \dots, M-1.$$

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Polynomial reproduction: if $Q^\ell(q) = q$, for any $q \in \mathbb{P}_{d_x, d_y}$ and $\ell = 0, \dots, M-1$, then $Q(q) = q$ for any $q \in \mathbb{P}_{d_x, d_y}$.

Hierarchical quasi-interpolants

The hierarchical QIs are a flexible tool:

- ▶ **properties of classic QIs** (based on local, not too expensive approximations, etc.);
- ▶ the adaptive nature of hierarchical splines allows to **increase the resolution of the mesh only when needed**;
- ▶ **the approximation order** is “inherited” by the tensor-product QIs employed.
- ▶ the coefficients can be computed “ignoring” the hierarchical structure: **we need to deal with B-splines only!**

Scattered data fitting

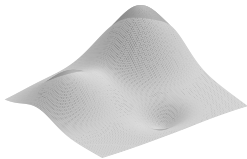
Scattered data fitting

$$F = \{(\mathbf{X}_i, \mathbf{f}_i), i = 1, \dots, n, \mathbf{X}_i \in \Omega \subset \mathbb{R}^2, \mathbf{f}_i \in \mathbb{R}^3, \mathbf{X}_i \neq \mathbf{X}_j \text{ if } i \neq j\}$$

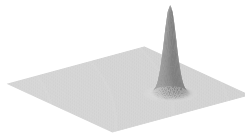
The goal is to construct a surface approximating the set of **scattered data** F .

We use the hierarchical splines to get a method **adapting**

- ▶ **to the shape of the data**
- ▶ **to the distribution of the data**



no sharp variations



sharp variations

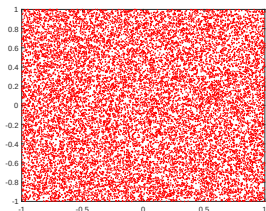
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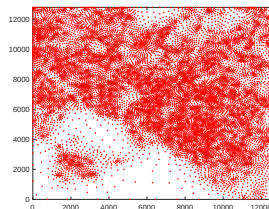
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“uniformly” scattered



“irregularly” scattered

Scattered data fitting

Scattered data fitting can be addressed with many approaches (RBFs, spline on triangulations, two-stage methods,...).

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We present a scheme based on THB-splines, focused on **the approximation of industrial data** (aircraft engine industry):

- ▶ **scanned data sets** generated by optical measurement systems
- ▶ **no artifacts** (oscillations) in the resulting model
- ▶ **high precision** of the approximation (10^{-4}m to 10^{-5}m tolerance for 85%-95% of the points)
- ▶ **compatibility** with CAD/CAE tools

Variable-degree local least squares QI

Given a tensor-product spline space V with basis $\mathcal{B} = \{B_J, J \in \Gamma\}$, we define the QI

$$Q(F) := \sum_{J \in \Gamma} \mu_J(F) B_J,$$

where $\mu_J(F) = (\mu_{J,1}(F), \mu_{J,2}(F), \mu_{J,3}(F)) \in \mathbb{R}^3$. Each $\mu_{J,i}(F)$ is the **coefficient of B_J in the polynomial solution**

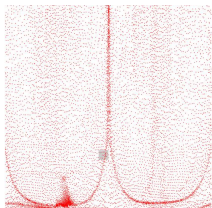
$$p_{J,i} = \sum_{K: \text{supp}(B_K) \cap \text{supp}(B_J) \neq \emptyset} c_{K,i} B_K \in \Pi_J,$$

of the **least squares problem on the local data set $F_J \subseteq F$**

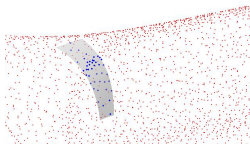
$$\min_{p \in \Pi_J} \sum_{h: (\mathbf{x}_h, f_{h,i}) \in F_J} (p(\mathbf{x}_h) - f_{h,i})^2,$$

where $\mathbf{f}_h = (f_{h,1}, f_{h,2}, f_{h,3})$, $h = 1, \dots, n$.

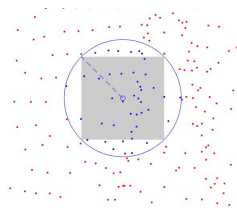
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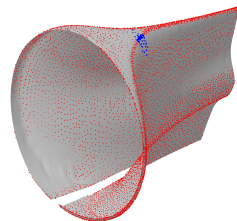
For each basis function...



...and define a polynomial approximation...



...we select a local set of data...



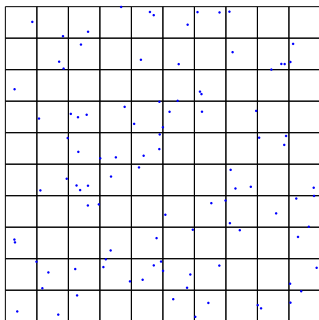
...from which we extract the coefficients used in the global approximation.

Variable-degree local least squares QI

STEP 1: SELECT F_J

$$F_J = \{(\mathbf{X}_i, \mathbf{f}_i) : \|\mathbf{X}_i - C_J\| \leq \rho_J\}$$

where C_J is the **center of $\text{supp}(B_J)$** and $\rho_J := (\text{diam}(\text{supp} B_J))/2$.
The radius ρ_J is increased until $|F_J| \geq 1$ (but $\rho_J \leq \rho_{J,\max}$).

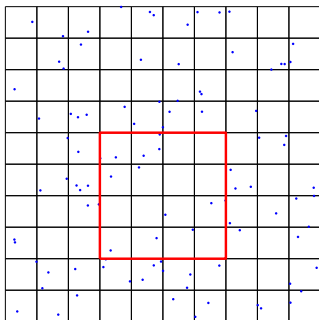


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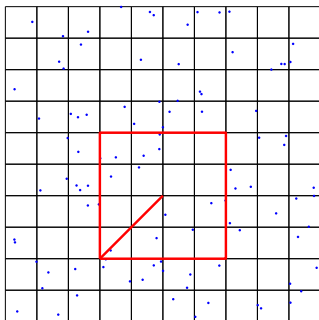


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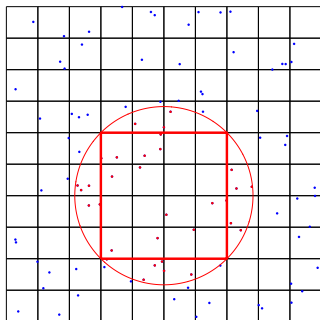


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Variable-degree local least squares QI

STEP 2: SELECT Π_J

- ▶ Π_J is initialized as the space of bivariate polynomials of total degree $d = \min\{d_x, d_y\}$;
- ▶ the degree of Π_J is iteratively decreased until both the following conditions are satisfied:
 - ▶ $|F_J| \geq \dim(\Pi_J)$;
 - ▶ the maximal singular value of the matrix giving the coefficient $\mu_J(F)$ is below a set threshold σ .

Scattered data hierarchical QI

We consider the **hierarchical version of the variable-degree local least squares QI**:

$$Q_{\mathcal{H}}(F) := \sum_{\ell=0}^{M-1} \sum_{J \in A_{\mathbf{d}}^{\ell}} \mu_J^{\ell} T_J^{\ell}$$

where $\mu_J^{\ell}(F) = (\mu_{J,1}^{\ell}(F), \mu_{J,2}^{\ell}(F), \mu_{J,3}^{\ell}(F)) \in \mathbb{R}^3$. Each $\mu_{J,i}^{\ell}(F)$ is the **coefficient of B_J^{ℓ} in the polynomial solution**

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of the **least squares problem on the local data set $F_J^{\ell} \subseteq F$**

$$\min_{p \in \Pi_J^{\ell}} \sum_{h: (\mathbf{x}_h, f_{h,i}) \in F_J^{\ell}} (p(\mathbf{x}_h) - f_{h,i})^2.$$

How is the hierarchical mesh constructed?

- ▶ **Initial mesh:** tensor-product mesh $\mathcal{G}_{\mathcal{H}}$ with $M = 1$ levels.

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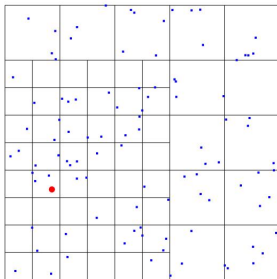
- ▶ **Initial mesh**: tensor-product mesh $\mathcal{G}_{\mathcal{H}}$ with $M = 1$ levels.
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- ▶ The goal is to have the **error** below a set tolerance **tolerance** ϵ in at least a percentage α of the data points.
- ▶ In order to **avoid refining too much** where there are few data points (which can lead to artifacts in the surface), we require a minimum number n_{loc} of **locally available data**.

How is the hierarchical mesh constructed?

- ▶ While the tolerance is not satisfied in the α percentage of the points (or the **maximum number of levels K** is reached), do:
 - ▶ for each $\mathbf{X}_i$ such that $e_i := \|Q_{\mathcal{H}}(F)(\mathbf{X}_i) - \mathbf{f}_i\|_2 > \epsilon$, mark the functions whose support contain it;
 - ▶ for each marked function, if its support contains n_{loc} points, its cells are refined.

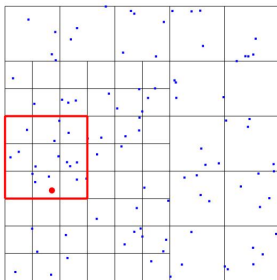


At $\mathbf{X}_i$ the tolerance is exceeded: the cells of all the supports containing it and containing $n_{loc} = 15$ points are subdivided ($d_x = d_y = 2$)

n_{loc} avoids to “over-refine” where there are too few data.

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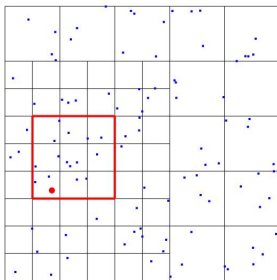


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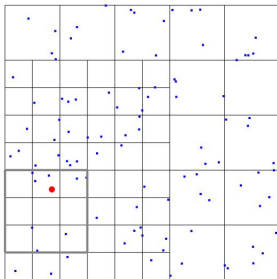


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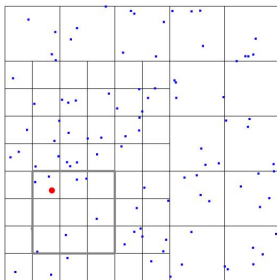


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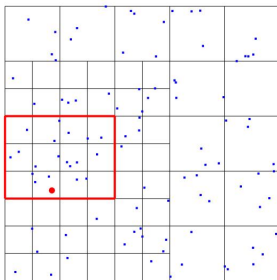


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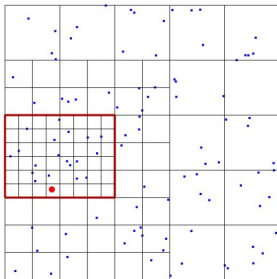


At $\mathbf{X}_i$ the tolerance is exceeded: the cells of all the supports containing it and containing $n_{loc} = 15$ points are subdivided ($d_x = d_y = 2$)

n_{loc} avoids to “over-refine” where there are too few data.

How is the hierarchical mesh constructed?

- ▶ While the tolerance is not satisfied in the α percentage of the points (or the **maximum number of levels K** is reached), do:
 - ▶ for each $\mathbf{X}_i$ such that $e_i := \|Q_{\mathcal{H}}(F)(\mathbf{X}_i) - \mathbf{f}_i\|_2 > \epsilon$, mark the functions whose support contain it;
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Error analysis

Definition (Buffa, Giannelli; 2016)

A hierarchical mesh $\mathcal{G}_{\mathcal{H}}$ is admissible of class m , $1 \leq m < M$, if the THB-splines taking non-zero values on any cell $c \in \mathcal{G}_{\mathcal{H}}$ belong to at most m successive levels.

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Theorem (B., Giannelli, Großmann, Sestini; 2017)

For any $f \in C^{d+1}(\Omega)$ ($d = \min\{d_x, d_y\}$) and any cell $c \in \mathcal{G}_{\mathcal{H}}$

$$\sup_{\mathbf{X} \in c} \|f(\mathbf{X}) - Q_{\mathcal{H}}(f)(\mathbf{X})\|_2 \leq (1 + \sigma) K D_f \left(\text{diam}(c) \right)^{d(c)+1}$$

- ▶ σ : bound for the maximum singular value of the local approximation matrix
- ▶ K : constant depending on degree and class of admissibility
- ▶ $d(c)$: minimum degree used in the polynomial approximations “active” on c
- ▶ D_f depends on the (local) derivatives of f of order $d(c)$

Numerical examples

Numerical examples

We compare the results obtained on some industrial data with those obtained with adaptive method based on **global regularized least squares approximation with THB-splines** introduced in [Kiss, Giannelli, Zore, Jüttler, Großmann, Barne; 2014].

Such method already proved that using THB-splines is an effective way to reduce the degrees of freedom. With this new algorithm our focus is addressing the **problem of unwanted oscillation**.

Numerical examples

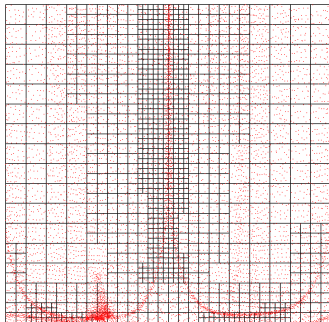
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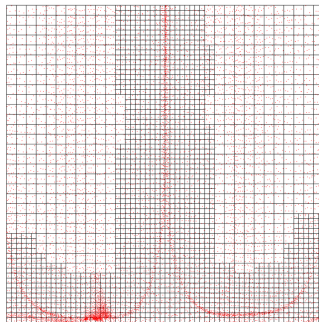
In all the tests, the parameters are chosen to have the **tolerance satisfied in at least $\alpha = 96\%$ of the data points**.

Numerical examples - tensile data

9281 data, tolerance $\epsilon = 5 \cdot 10^{-5}$, $\mathbf{d} = (2, 2)$



Global regularized least square fitting with 4 levels and smoothing coefficient $\lambda = 10^{-6}$ (DOF=1136)



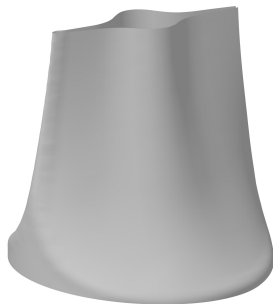
Fitting based on variable degree local least squares with 4 levels, $n_{loc} = 20$ (DOF=2506)

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9281 data, tolerance $\epsilon = 5 \cdot 10^{-5}$, $\mathbf{d} = (2, 2)$



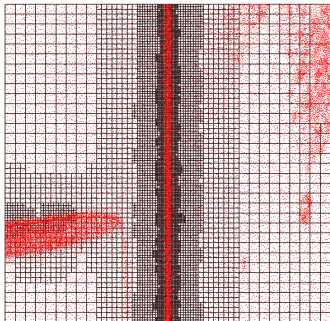
Global regularized least square fitting
with 4 levels and smoothing coefficient
 $\lambda = 10^{-6}$ (DOF=1136)



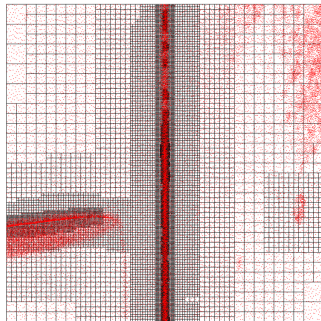
Fitting based on variable degree local
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Numerical examples - blade data

27191 data, tolerance $\epsilon = 2 \cdot 10^{-5}$, $\mathbf{d} = (3, 3)$



Global regularized least square fitting with 6 levels and smoothing coefficient $\lambda = 10^{-8}$ (DOF=8164)



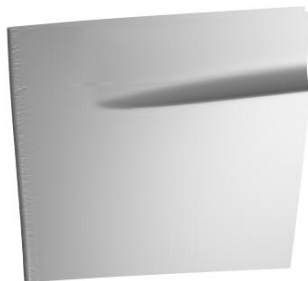
Fitting based on variable degree local least squares with 7 levels, $n_{loc} = 60$ (DOF=12721)

Numerical examples - blade data

27191 data, tolerance $\epsilon = 2 \cdot 10^{-5}$, $\mathbf{d} = (3, 3)$



Global regularized least square fitting with 6 levels and smoothing coefficient $\lambda = 10^{-8}$ (DOF=8164)



Fitting based on variable degree local least squares with 7 levels, $n_{loc} = 60$ (DOF=12721)

Improving the method?

Refinement

- ▶ While the tolerance is not satisfied in the α percentage of the points (or the maximum number of levels K is reached), do:
 - ▶ for each $\mathbf{X}_i$ such that $e_i := \|Q_{\mathcal{H}}(F)(\mathbf{X}_i) - \mathbf{f}_i\|_2 > \epsilon$, mark the functions whose support contain it;
 - ▶ for each marked function, if its support contains n_{loc} points, its cells are refined.

Local data set

$$F_J = \{(\mathbf{X}_i, \mathbf{f}_i) : \|\mathbf{X}_i - C_J\| \leq \rho_J\} \quad (\text{circle around the support})$$

Improving the method?

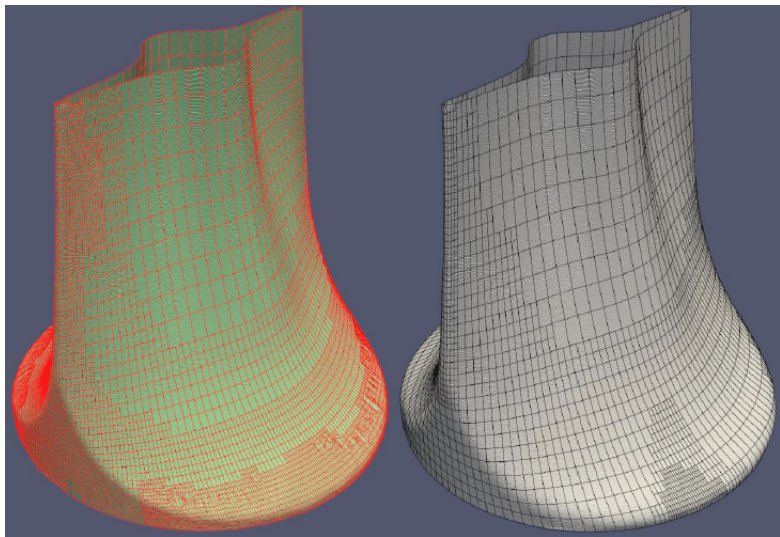
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- ▶ While the tolerance is not satisfied in the α percentage of the points (or the maximum number of levels K is reached), do:
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 - ▶ for each marked function, if the supports of its children functions all contain n_{loc} points, the cells in its support are refined.

Local data set

$$F_J = \{(\mathbf{X}_i, \mathbf{f}_i) : \mathbf{X}_i \in R_J := \overline{\text{supp}(B_J)}\} \quad (\text{support}).$$

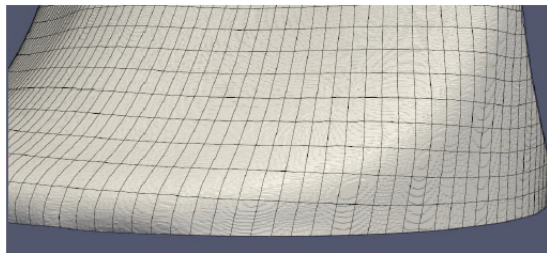
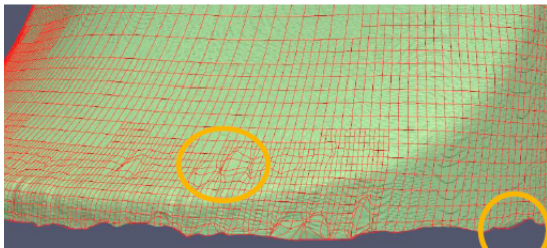
Numerical examples - new strategy



Old strategy (left) vs new strategy (right)

REDUCED ACCURACY (96% vs 87% of points within tolerance)...

Numerical examples - new strategy



Old strategy (top) vs new strategy (bottom)
...BUT NO OSCILLATIONS!

Remarks...

- ▶ **Local adaptive** scattered data fitting algorithm which can be integrated with existing CAD-CAE tools.
- ▶ The adaptivity provided by the hierarchical spaces is not enough: the **adaptivity of the local approximants is crucial**.
- ▶ The **refinement strategy** takes in account not only **local error**, but also **local number of data**.
- ▶ In this industrial setting, we privilege **the quality of the approximation (NO oscillations)**.

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...and new developments

- ▶ Very promising results with improved refinement strategy (**most of remaining oscillations removed**).
- ▶ Improvements for **special types of scattered data**?
- ▶ **Other local approximations**?

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Thank you for your attention!